Answer Key – Team Round (10 marks each)

1. **₹450**. 100 bookmarks for ₹750, so ₹7.50 each. Kabir: 60 × 7.50 = 450.

2. **384 cm²**. Side = 8 cm ($8^3 = 512$). Surface area = $6 \times 8^2 = 384$.

3. **8**. Draws give 7 points, so wins give 45 points, which is 15 wins. Losses = 30 − 15 − 7 = 8.

4. **25**. $12^2 + 13^2 = 144 + 169 = 313$, so 12 + 13 = 25.

5. **36**. After 6 men leave, 48 people remain in the ratio 1 : 3, so 12 men and 36 women.

6. **Day 17**. After 16 days and nights it is at 16 m. On day 17 it climbs 4 m and reaches 20 m before it can slip back.

7. **3 hours 26 minutes**. Per hour: 1/4 + 1/6 − 1/8 = (6 + 4 − 3)/24 = 7/24 of the tank. Time = 24/7 h = 3 h 25 5⁄7 min ≈ 3 h 26 min.

8. **16**. $12.5n + 38 = 14(n + 1)$, so $1.5n = 24$ and $n = 16$.

9. **10**. Add 1 to both sides: $(a + 1)(b + 1) = 35 = 5 \times 7$, so $a = 4$ and $b = 6$. (1×35 would make $a = 0$, which is not positive.)

10. **112 cm²**. Each quarter circle is $1/4 \times 22/7 \times 14^2 = 154$. The leaf is two segments, each a quarter circle minus the triangle (½ × 14 × 14 = 98): 2 × (154 − 98) = 112.

Answer Key – Individual Round (+4 / −1)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	6	B	11	A	16	B	21	C	26	A
2	B	7	A	12	E	17	C	22	E	27	D
3	C	8	D	13	A	18	A	23	C	28	E
4	E	9	E	14	B	19	D	24	D	29	B
5	C	10	C	15	D	20	B	25	E	30	A

Individual Round – Worked Solutions

1. D (30). 2 ★ 3 = 4 + 6 = 10. 10 ★ 1 = 20 + 10 = 30.

2. B (9). 18, 27, 36, 45, 54, 63, 72, 81, 90.

3. C (7:20 pm). 4:45 + 2 h = 6:45; + 35 min = 7:20.

4. E (150). 15% of 300 = 45, and 30% of 150 = 45.

5. C (3). $2^4 = 4^2$, $4^3 = 8^2$, $5^6 = 125^2$. 3^5 and 6^5 have odd powers, so they are not squares.

6. B (21). Average = 18, so the numbers are 15, 17, 19, 21.

7. A (5/8). 3/5 = 0.6 and 2/3 ≈ 0.667. 5/8 = 0.625 lies between them. The others are 0.583, 0.6875, 0.571 and 0.7.

8. D (9). $4x - x = 27$, so $x = 9$.

9. E (64). Each fold doubles the layers: $2^6 = 64$.

10. C (10). 15 boys and 25 girls; 25 − 15 = 10.

11. A (26). The total rises by 2 + 4 + 6 + 8 + 10 = 30, so the mean rises by 30 ÷ 5 = 6.

12. E (100°). The other two angles total 140° = 7 parts of 20°, so they are 40° and 100°.

13. A (7). Units digits of powers of 7 repeat 7, 9, 3, 1. 2025 = 4 × 506 + 1, so the units digit is 7.

14. B (96%). 1.2 × 0.8 = 0.96.

15. D (60 cm). The square has side 9 and area 81. The rectangle is 3 × 27, so its perimeter is 2(3 + 27) = 60.

16. B (Monday). Saturdays alternate between even and odd dates. Three even ones require Saturdays on 2, 9, 16, 23, 30. The 23rd is a Saturday, so the 25th is a Monday.

17. C (15). $2025 = 3^4 \times 5^2$, which has (4 + 1)(2 + 1) = 15 factors.

18. A. If I is true, II is false, so Bina is the tallest, which contradicts I. If only II is true, Chetan must be the tallest but III being false makes him the shortest, a contradiction. So only III is true: Bina is the tallest, then Chetan, then Asha.

19. D (12). 6 melons = 8 oranges = 12 apples.

20. B (2). 2400 mL × 3 s = 7200 s = 2 hours.

21. C (60 km/h). The bike gains 30 km in 1.5 h, so it is 20 km/h faster than the car: 40 + 20 = 60.

22. E (36 cm). Pushing the step outwards gives a 10 × 8 rectangle, so the perimeter is 2(10 + 8) = 36.

23. C (150°). Angles around point *A*: 360° − 120° (hexagon) − 90° (square) = 150°.

24. D (9). 4!/2! = 12 arrangements of 2, 2, 0, 5. Those starting with 0 number 3!/2! = 3, leaving 12 − 3 = 9.

25. E (12). One edge cube (not a corner) on each of the 12 edges.

26. A (18). The cube has 12 edges, and each cut adds 3 new ones: 12 + 3 + 3 = 18.

27. D (20). Choose any 3 of the 6 points: (6 × 5 × 4) ÷ (3 × 2 × 1) = 20.

28. E (+1). The first column gives the magic sum: −3 + 4 − 1 = 0. One diagonal gives the centre: −3 + centre + 3 = 0, so the centre is 0. The other diagonal: Y + 0 − 1 = 0, so Y = 1.

29. B (4). Dev–Chetan–Esha (in either order) form a block with Chetan in the middle. With Asha in seat 1, the block fills seats 2–4 (Bina in seat 5) or seats 3–5 (Bina in seat 2). That is 2 positions × 2 orders = 4.

30. A (41). Multiples of 3: 33. Multiples of 5: 20. Multiples of both (15): 6. Answer: 33 + 20 − 2 × 6 = 41.